

GANESH VELURU

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Education

Master of Science in Computer Science, University of South Florida, Tampa, FL | GPA: 4.00/4.00 | Aug 2023 – May 2025

Bachelor of Technology in Computer Science, KITSW, Warangal, India | GPA: 9.75/10.00 | July 2018 – May 2022

Technical Skills

- **Networking & Protocols:** TCP/IP, Subnetting (VLSM/CIDR), VLANs, Spanning Tree Protocol (STP), Ethernet, NAT, PAT, MPLS, DNS, TLS, DHCP, TFTP, SFTP, IPv4, IPv6, Telnet.
- **Routing & Switching:** Inter-VLAN Routing, Layer 2/3 Switching, OSPF, BGP, EIGRP, RIP.
- **Operating Systems & Virtualization:** Windows (10/8/7/XP, Server 2008/2003), Mac OS, Linux, Unix.
- **Cloud Networking:** AWS VPC, EC2 Networking, Route Tables, Route53, Direct Connect, Elastic Load Balancing, Auto scaling.
- **Tools & Platforms:** Cisco Packet Tracer, GNS3, Wireshark, Putty.
- **Automation & Scripting:** Python (Netmiko), Shell Scripting, Bash, Paramiko, Ansible, RESTful APIs.
- **Network Security:** Cisco ESA (Email Security Appliance), Phishing Detection, SPF/DKIM/DMARC Enforcement, Email Routing & Content filtering, SSH, ACLs, Threat Analysis, VPNs, Firewall Policy Implementation, Vulnerability Assessment.
- **Monitoring & Troubleshooting:** Ping, Traceroute, syslog, SNMP, dig, curl, Wireshark, Postman.
- **Additional:** C, Java, Python, SQL, MySQL, PostgreSQL, MongoDB, Web Technologies (HTML, CSS, JavaScript, PHP, Bootstrap), Machine Learning, Docker, Git, Microsoft Office.

Experience

Network Engineer

Cisco Systems – Bangalore, India

March 2022 – August 2023

- Expertly analyzed and resolved complex email security threats—including phishing, malware, spam, and spoofing—using Cisco Email Security Appliance (ESA) and threat intelligence feeds, improving email delivery security for enterprise clients
- Conducted deep packet inspection and flow analysis on email traffic to identify vulnerabilities and ensure secure mail flow, maintaining 98%+ customer satisfaction ratings.
- Configured and enforced granular email security policies (sender authentication, content filtering, reputation-based blocking) that reduced phishing incidents within client environments.
- Automated alerting workflows by developing Python-based bots that monitored orphaned customer requests and escalated unresolved issues within 5 minutes, increasing TAC response efficiency.
- Collaborated with cross-functional teams and assisted customers in deploying and tuning secure email appliances, integrating LDAP, SMTP, and TLS/SSL, while documenting recurring threats to enhance internal knowledge and improve incident response.
- Achieved Bronze Medal for driving 10+ qualified technical sales leads by combining technical expertise and customer engagement.

Software Development Intern

Cisco Systems – Bangalore, India

September 2021 – March 2022

- Completed comprehensive training and earned certifications in CCNA and Cisco DevNet, strengthening foundational knowledge in network infrastructure, protocols, and automation.
- Designed and developed a full-stack network diagnostic parser tool using Python Flask, HTML, CSS, JavaScript, Bootstrap, and SQL to streamline troubleshooting of MAC and Linux diagnostic bundles.
- Enabled engineers and customers to upload network diagnostic bundles and parse critical configuration and log data, improving root cause analysis speed and reducing troubleshooting time by 75%.
- Automated parsing and visualization of network diagnostic files by applying networking concepts and programming skills, enabling faster detection of issues and bridging gaps between network operations and support tools.

AIML Intern

Smart Bridge – Hyderabad, India

May 2021 – August 2021

- Completed an AI/ML externship in collaboration with IBM, gaining hands-on experience through guided projects and e-learning courses on foundational machine learning concepts.

Projects

Enterprise Network Design & Implementation

Cisco Packet Tracer | Networking, Routing, Switching, Security

- Designed a redundant enterprise network for 600+ users using multilayer switches and dual routers in a hierarchical model.
- Configured VLANs, DHCP, PAT, inter-VLAN routing, and OSPF for dynamic route advertisement.
- Implemented dual ISP connectivity, tested failover, and used ACLs for traffic filtering.
- Secured the network using SSH, console passwords, and port security with sticky MAC for Finance VLAN.
- Simulated and verified all configurations in Cisco Packet Tracer, including wireless setup and IP subnetting.

xv6 kernel Development & Security Internals (MIT Unix-like OS)

- Implemented custom system calls, modified process scheduler, added command-line support to exec (), and developed a secure find utility in C. Explored memory isolation, virtual memory, and kernel-level auditing concepts.

Network Config Management and Drift Detection Automation

Python, Netmiko, Git, YAML, SSH, diffli

- Automated retrieval of running config via SSH for 20+ network devices using Netmiko, enabling consistent and reliable backups.
- Integrated Git for version-controlled storage and history tracking of device configurations.
- Automated config drift detection by comparing current vs. previous backups, committing only on meaningful changes.
- Used YAML for dynamic, scalable device inventory and configuration management.
- Improved network security posture by enabling early detection of unauthorized or unintended configuration changes.

Certifications

- Cisco Certified Network Associate (CCNA), Cisco Certified DevNet Associate
- AWS Certified Solutions Architect, AWS Certified Cloud Practitioner
- Infosys Certified Software Programmer
- Programming in Java, Programming in C – IIT Kharagpur
- Java Programming, Database Design and Programming with SQL, AIML – Oracle
- Python 3 Programming – University of Michigan
- SQL for Data Analytics – Udemy